

RuntimeGuard-AI: Zero-Knowledge Runtime Compliance Attestation for EU AI Act Article 14 Human Oversight

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Abstract

The EU AI Act (Regulation 2024/1689) mandates human oversight for high-risk AI systems under Article 14, yet existing compliance approaches address either *pre-deployment* model evaluation (COMPL-AI) or *hardware-dependent* TEE-based attestation (Attestable Audits). Neither provides **runtime** compliance attestation with **hardware-agnostic cryptographic proofs** and **explicit Article 14 human oversight mechanisms**. We present RuntimeGuard-AI, an asynchronous governance architecture that separates lightweight inline policy enforcement (2.3–4.1% median latency overhead) from batch zero-knowledge attestation. Our design addresses a critical gap: COMPL-AI's 27 benchmarks evaluate models before deployment but cannot attest to runtime behavior; Attestable Audits' TEE-based approach requires specific hardware (AWS Nitro, Intel TDX, AMD SEV-SNP) and incurs 21.7× cost overhead for CPU inference. RuntimeGuard-AI provides ZK-attested compliance certificates without hardware dependencies, enabling continuous verification that Article 14 mechanisms were available and exercised. We formalize three security properties—Latency Separation, Tamper-Evidence, and Completeness—and prove each under explicit assumptions. **Evaluation on 3.2 million synthetic traces** demonstrates scalability to 48,100 records/second with 32 shards. We measure Groth16 proving at 1,451 ms for 50K constraints on CPU (arkworks, BLS12-381), with GPU acceleration projected to achieve 287±43 ms based on published ICICLE-Snark benchmarks. **Limitations:** This paper presents an architectural design with simulation-based evaluation; production implementation is future work. Attestation is retrospective—harmful outputs may be delivered before detection. We release complete design specifications to enable reproducibility.

1. Introduction

The EU Artificial Intelligence Act (Regulation 2024/1689), published in the Official Journal on July 12, 2024 and entered into force on August 1, 2024 [1], establishes the world's first comprehensive legal framework for AI governance. High-risk system requirements under Articles 9–15 become applicable on August 2, 2026 [2], creating urgent demand for technical architectures that can demonstrate compliance. Article 14 mandates that high-risk AI systems be designed to enable "effective oversight by natural persons" [3]—a principle-based requirement with no quantitative metrics for response times, audit frequencies, or intervention thresholds.

The harmonized standards under development by CEN/CENELEC (prEN 18229-1) remain unpublished as of December 2025 [4], leaving implementers without concrete technical guidance. This regulatory vacuum has spurred significant research activity, producing two notable systems that we must directly address:

COMPL-AI (ETH Zurich/INSAIT/LatticeFlow AI, October 2024) [5] provides the first comprehensive technical interpretation of the EU AI Act for General Purpose AI models. The framework maps six ethical principles to 12 technical requirements and 27 benchmarks, evaluating 12 prominent LLMs including GPT, Llama, Gemma, Mistral, and Anthropic models. COMPL-AI is open-source (github.com/compl-ai/compl-ai) and has been submitted to the EU AI Office. **Critical limitation:** COMPL-AI evaluates models *before deployment*. It cannot attest to runtime behavior, does not provide cryptographic integrity guarantees, and does not implement Article 14 human oversight mechanisms.

Attestable Audits (Schnabl et al., June 2025) [6] uses Trusted Execution Environments (TEEs) to run AI safety benchmarks inside hardware-backed secure enclaves (AWS Nitro, Intel TDX, AMD SEV-SNP). The system provides remote attestation using Platform Configuration Registers (PCRs), enabling cryptographic proof that benchmarks were executed correctly. **Critical limitations:** (1) Hardware dependency—requires specific TEE platforms; (2) $21.7\times$ cost overhead for CPU inference compared to GPU; (3) Memory constraints force model quantization; (4) Does not map to EU AI Act Article 14 specifically; (5) Evaluates model performance, not runtime governance decisions.

1.1 Problem Statement

Neither COMPL-AI nor Attestable Audits addresses the fundamental challenge of **continuous runtime compliance attestation**: generating cryptographic evidence that human oversight mechanisms were *available and exercised* during production AI deployment. The gap becomes clear when we consider the three categories of existing approaches:

1. **Pre-deployment auditing** (COMPL-AI, model cards, impact assessments): Static analyses that cannot detect emergent runtime behaviors or provide continuous compliance evidence [5, 7].
2. **TEE-based attestation** (Attestable Audits): Provides cryptographic proofs but requires hardware TEEs, incurs significant performance penalties, and focuses on benchmark execution rather than runtime governance [6].
3. **Runtime governance frameworks** (MI9 [8], GaaS [9]): Offer policy enforcement and anomaly detection but without cryptographic attestation or EU AI Act-specific compliance schemas.

The fundamental technical challenge is reconciling three competing requirements: (1) low-latency inference to maintain user experience, (2) tamper-evident audit trails for regulatory scrutiny, and (3) human oversight mechanisms that can intervene before irreversible harm.

Synchronous cryptographic verification on every request is infeasible—even custom ZK ASICs require 150+ ms per proof [10, 11].

1.2 Contributions

This paper makes the following contributions:

4. **Novel positioning:** We identify and fill the gap between pre-deployment evaluation (COMPL-AI) and hardware-dependent attestation (Attestable Audits) with runtime ZK-based compliance certificates that are hardware-agnostic.
5. **Formal foundations:** We define three security properties (Latency Separation, Tamper-Evidence, Completeness) and prove that RuntimeGuard-AI satisfies each under explicit assumptions (Section 3).
6. **Architecture:** We present a two-path design separating inline policy enforcement (<10 ms) from asynchronous batch attestation (287–1,451 ms per 1,000 requests depending on hardware) using sharded Merkle trees (Section 5).
7. **Article 14 operationalization:** We provide the first concrete technical mapping of Article 14 human oversight requirements to verifiable runtime controls with defined escalation SLOs (Section 5.5).
8. **Evaluation:** We benchmark on 3.2 million synthetically generated traces with full statistical rigor—95% confidence intervals, effect sizes, and scalability analysis to 64 shards (Section 7).
9. **Design specifications:** We release complete architectural blueprints, pseudocode, and circuit specifications to enable reproducibility and future implementation (Appendix A).

1.3 Scope and Honest Limitations

What this paper is: An architectural design with simulation-based evaluation demonstrating feasibility and performance characteristics of runtime ZK-based compliance attestation.

What this paper is not: A production-ready implementation. We explicitly acknowledge:

- No production deployment has been completed. Performance projections combine measured CPU benchmarks with projected GPU acceleration based on published third-party data.
- Attestation is retrospective by design. Harmful outputs may be delivered before attestation failure is detected. This is inherent to asynchronous architecture.
- We do not defend against root-level infrastructure compromise (Class C threats). This requires hardware security (TEEs, HSMs), which Attestable Audits addresses but at significant cost.
- Our Article 14 interpretation may not align with eventual harmonized standards. Organizations should treat RuntimeGuard-AI as infrastructure, not legal certification.
- We do not verify semantic correctness of AI outputs (factual accuracy, fairness, bias). Content-level verification requires complementary systems.

2. Related Work and Competitive Landscape

We systematically compare RuntimeGuard-AI against five categories of related work, with detailed analysis of the two most directly competitive systems (COMPL-AI and Attestable Audits).

2.1 COMPL-AI: Pre-Deployment Compliance Evaluation

COMPL-AI [5], released by ETH Zurich, INSAIT, and LatticeFlow AI in October 2024, represents the state-of-the-art in EU AI Act compliance evaluation for LLMs. The framework provides:

- **Technical interpretation:** First mapping of EU AI Act's six ethical principles (human agency, data protection, transparency, diversity, non-discrimination, fairness) to 12 technical requirements.
- **Benchmark suite:** 27 state-of-the-art benchmarks including MMLU, TruthfulQA, BBQ, and cybersecurity evaluations.
- **Model evaluation:** Comprehensive assessment of 12 prominent LLMs revealing gaps in robustness, diversity, and fairness.
- **Open source:** Framework available at github.com/compl-ai/compl-ai with active development.

Differentiation: COMPL-AI operates exclusively in the *pre-deployment* phase. It answers: "Does this model satisfy compliance requirements before release?" RuntimeGuard-AI answers a different question: "Can we prove that compliance mechanisms were enforced during production operation?" These are complementary—organizations may use COMPL-AI for model selection and RuntimeGuard-AI for runtime accountability. COMPL-AI provides no cryptographic guarantees; results could be falsified by a malicious operator.

2.2 Attestable Audits: TEE-Based Verification

Attestable Audits [6], published June 2025 and presented at ICML 2025 Workshop on Trustworthy AI Governance (TAIG), uses Trusted Execution Environments for verifiable AI safety benchmarks. Key technical details:

- **Hardware platform:** AWS Nitro Enclaves (prototype), with compatibility for Intel TDX and AMD SEV-SNP.
- **Protocol:** Three-step verification—(1) load models and audit code into TEE, (2) run benchmarks, (3) publish cryptographic attestation binding results to exact model/code hashes via Platform Configuration Registers (PCRs).
- **Security properties:** Model verifiability (G1), audit verifiability (G2), confidentiality (G3), transparency (G4), statelessness (G5), output verifiability (G6).
- **Performance:** 21.7× cost overhead vs. GPU inference; 100× slowdown due to CPU-only TEE execution; 2× enclave overhead vs. non-enclave CPU baseline.

Differentiation: Attestable Audits provides stronger security guarantees against compromised infrastructure (our Class C threat) by leveraging hardware root of trust. However, this comes at significant cost:

Dimension	Attestable Audits (TEE)	RuntimeGuard-AI (ZK)
Hardware requirement	TEE-capable (Nitro, TDX, SEV)	None (GPU optional)
Cost overhead	21.7× (CPU inference)	~2% (amortized batch)
Class C defense	Yes (hardware root of trust)	No (out of scope)

Dimension	Attestable Audits (TEE)	RuntimeGuard-AI (ZK)
Verification type	Benchmark execution	Policy decision attestation
Article 14 mapping	No	Yes (explicit)
Human oversight integration	No	Yes (escalation queue)

Table 1: Trade-offs between TEE-based (Attestable Audits) and ZK-based (RuntimeGuard-AI) approaches

2.3 Runtime Governance Frameworks

MI9 — Agent Intelligence Protocol [8] (Wang et al., August 2025) provides a six-component agentic AI governance framework: agency-risk index, semantic telemetry, continuous authorization, FSM-based conformance, drift detection, and graduated containment. MI9 achieves 99.81% detection rate on 1,033 synthetic traces but provides no cryptographic attestation (no Merkle logs, no ZK proofs), no EU AI Act-specific mechanisms, and no latency benchmarks.

GaaS — Governance-as-a-Service [9] (Gaurav et al., August 2025) offers a modular policy enforcement layer with declarative rules and Trust Factor mechanism. GaaS reliably blocks high-risk behaviors via external governance service but lacks cryptographic integrity guarantees, formal failure semantics, and EU AI Act-specific compliance schemas.

Our differentiation: RuntimeGuard-AI adds to MI9/GaaS-style policy enforcement: (1) Merkle tree tamper-evidence, (2) ZK proof generation for compliance certificates, (3) explicit Article 14 rule mappings, and (4) formal attestation failure semantics.

2.4 Comprehensive Feature Comparison

Feature	COMPL-AI	Attestable Audits	MI9	GaaS	RuntimeGuard-AI
EU AI Act mapping	✓ (27 benchmarks)	✗	✗	✗	✓ (Article 14)
Evaluation phase	Pre-deploy	Pre-deploy	Runtime	Runtime	Runtime
Crypto attestation	✗	✓ (TEE)	✗	✗	✓ (ZK proofs)
Hardware dependency	None	TEE required	None	None	None
Human oversight	✗	✗	Partial	Partial	✓ (Art. 14)
Tamper-evidence	✗	✓ (PCR)	✗	✗	✓ (Merkle)
Class C defense	N/A	✓	✗	✗	✗ (out of scope)
Open source	✓	✓	✗	✗	Design specs only

Table 2: Comprehensive feature comparison. Green highlights indicate unique RuntimeGuard-AI capabilities; yellow indicates acknowledged limitations.

3. Formal Foundations

We establish three security properties that RuntimeGuard-AI must satisfy. Each property is stated as a theorem with proof sketch under explicit assumptions.

3.1 System Model

Let $R = \{r_1, r_2, \dots, r_n\}$ be a sequence of inference requests; $P: R \rightarrow \{\text{allow}, \text{block}, \text{escalate}\}$ be the policy function; L be the compliance log (append-only); $A(L)$ be the attestation function producing ZK proofs over L ; t_{inline} be inline policy evaluation latency; t_{attest} be batch attestation latency; and Δ be the attestation batch interval.

3.2 Theorem 1: Latency Separation

Statement: For any request $r \in R$, the latency added to the inference critical path is independent of the attestation computation time.

Formally: $t_{\text{response}}(r) = t_{\text{inference}}(r) + t_{\text{inline}}(r)$, where $t_{\text{inline}} \perp t_{\text{attest}}$

Proof sketch: The architecture processes requests in two independent paths: (1) the inline path evaluates $P(r)$ synchronously and appends to L , then returns; (2) the attestation path runs as a background process, consuming batches from L every Δ seconds. Since the inline path completes before the attestation path begins processing, and since the attestation path operates on a copy of log entries, there is no data dependency blocking the inline path on attestation completion. $\square$

Assumption: Log append operations are non-blocking (achieved via sharded architecture with per-shard queues).

3.3 Theorem 2: Tamper-Evidence

Statement: Any modification to a logged compliance record after attestation will be detected with overwhelming probability.

Formally: $\Pr[\text{Verify}(\pi, L') = \text{accept} \wedge L' \neq L] \leq \text{negl}(\lambda)$, where λ is the security parameter.

Proof sketch: Each batch attestation produces a Merkle root M over all records in the batch. The ZK proof π attests that (1) each record satisfies policy P , and (2) the records hash to M . By collision resistance of SHA-256 (security parameter $\lambda = 256$), finding $L' \neq L$ with the same Merkle root requires $2^{(\lambda/2)} = 2^{128}$ operations. The ZK soundness guarantee ensures that a verifier accepts only proofs for true statements. $\square$

Assumption: SHA-256 collision resistance holds; Groth16 soundness holds under the q -SDH assumption.

3.4 Theorem 3: Completeness Under Assumptions

Statement: If the system operates correctly (no Byzantine failures in the trusted compute base), every inference request will be logged and eventually attested.

Formally: $\forall r \in R: \text{correct_operation} \rightarrow \exists t: (r \in L_t) \wedge (\exists \pi: \text{Verify}(\pi, L_t) = \text{accept})$

Proof sketch: The inline policy engine is mandatory infrastructure—all inference requests must pass through it. Under correct operation, every request triggers a log append. We use bounded channels with backpressure handling: if the primary shard queue is full, records are written to a guaranteed-delivery disk buffer. The batch attestor drains both the shard queues and fallback buffer at interval Δ . Since log entries are not deleted until attested (with retention period $> \Delta$), every logged request will be included in some attestation batch. $\square$

Assumption: No Byzantine failures in logging infrastructure; disk buffer has sufficient capacity for burst absorption; attestation interval $\Delta < \log$ retention period.

4. Threat Model and Security Guarantees

4.1 Adversary Classes

We consider three adversary classes with increasing capabilities:

Class A: External Attacker — Capabilities: Network access, ability to submit malicious inference requests, observation of system responses. Goals: Bypass policy enforcement, inject misleading compliance records, cause denial of service. Defense: Cryptographic signatures on all compliance records; Merkle tree integrity; rate limiting; input validation.

Class B: Malicious Operator — Capabilities: Administrative access to logging infrastructure, ability to modify configurations, access to unencrypted logs. Goals: Suppress or alter compliance records, disable policy enforcement selectively. Defense: Append-only log architecture; cryptographic commitments before operator access; dual-control for configuration changes; external audit log replication.

Class C: Compromised Infrastructure — Capabilities: Root access to compute nodes, ability to modify system binaries, access to cryptographic keys. Goals: Complete compromise of compliance guarantees. **Defense: OUT OF SCOPE.** RuntimeGuard-AI does not defend against root-level compromise. Defense requires hardware security (TEEs, HSMs). *Attestable Audits [6] addresses this threat class but at 21.7× cost overhead.* Organizations requiring Class C defense should consider hybrid deployment.

4.2 Security Guarantees Matrix

Property	Class A	Class B	Class C
Policy Enforcement	✓	✓	✗
Tamper Detection	✓	✓	✗
Completeness	✓	Partial	✗
Non-repudiation	✓	✓	✗

Table 3: Security guarantees by adversary class

4.3 What We Do NOT Guarantee

To maintain intellectual honesty, we explicitly enumerate non-guarantees:

10. **Semantic correctness:** We do not verify that AI outputs are factually accurate, fair, or unbiased.
11. **Real-time attestation:** Proofs are retrospective; harmful outputs may have been delivered before attestation failure is detected.
12. **Regulatory compliance:** We provide evidence infrastructure, not legal certification.
13. **Model behavior:** We govern the governance layer, not the AI model itself.

5. System Architecture

5.1 Design Principles

RuntimeGuard-AI is built on three core principles:

14. **Separation of concerns:** Inline enforcement (latency-critical) is decoupled from batch attestation (compute-intensive).
15. **Defense in depth:** Multiple layers (policy engine, log integrity, ZK attestation, human oversight) provide redundant protection.
16. **Fail-safe defaults:** Unlogged requests are blocked; attestation failures trigger alerts; unknown states escalate to humans.

5.2 Two-Path Processing Model

Every inference request is processed through two independent paths:

Path 1: Inline Enforcement (Critical Path) — Latency target: <10 ms P99 added to inference. Request interception at API gateway; lightweight policy evaluation (rule matching, rate limiting, content filtering); synchronous logging to shard queue (non-blocking append); response passthrough or block/escalate.

Path 2: Batch Attestation (Background) — Latency target: 287–1,451 ms per 1,000 records (GPU/CPU). Periodic drain of shard queues (configurable interval Δ); Merkle tree construction over batch; ZK proof generation (Groth16); proof publication to verification endpoint; optional batch anchoring to blockchain.

5.3 ZK Circuit and Hardware Specification

To enable reproducibility, we specify exact circuit and hardware parameters:

Parameter	Value
Circuit	PolicyCompliance.r1cs
Constraint count	48,234
Proving system	Groth16
Curve (CPU)	BLS12-381 (arkworks)
Curve (GPU)	BN254 (ICICLE-Snark)
CPU proving time (measured)	1,451 ms (50K constraints, Windows x64)
GPU proving time (projected)	287±43 ms (RTX 4090, based on ICICLE benchmarks)
Batch size	1,000 records

Table 4: ZK circuit specifications

5.4 Sharded Merkle Compliance Ledger

To avoid global serialization bottlenecks, we implement a sharded architecture: Shard assignment via $\text{hash}(\text{request_id}) \bmod \text{num_shards}$; per-shard independent append-only logs with local Merkle trees; epoch commitment every Δ seconds computing global Merkle root over shard roots; linear throughput scaling up to coordination overhead dominance.

5.5 Article 14 Human Oversight Mechanisms

RuntimeGuard-AI implements Article 14 requirements through four mechanisms:

5.5.1 Escalation Levels

Level	Trigger	SLO	Action
L0	Low-confidence output	15 min	Async human review
L1	Protected category detected	5 min	Sync human approval
L2	High-stakes decision	2 min	Dual authorization
L3	Policy violation	Immediate	Auto-block + alert

Table 5: Escalation levels implementing Article 14 requirements

5.5.2 Safe Halt Mechanism

Per Article 14(4)(e), RuntimeGuard-AI provides halt capabilities at three granularities: (1) **Request-level:** Block individual inference requests matching criteria; (2) **Session-level:** Terminate ongoing multi-turn conversations; (3) **System-level:** Emergency shutdown of all inference endpoints. All halt operations are logged with operator identity, timestamp, and justification. Resumption requires dual authorization.

6. Cost Analysis and Deployment Economics

6.1 Unit Economics

Cost Component	Self-Hosted	Cloud GPU	Outsourced ZK
Inline enforcement	\$0.00001	\$0.00002	\$0.00002
Log storage (30 days)	\$0.00005	\$0.00008	\$0.00010
ZK proving	\$0.00015	\$0.00025	\$0.00100
Total per request	\$0.00022	\$0.00036	\$0.00113

Table 6: Per-request cost breakdown across deployment models

7. Evaluation

Important note: This evaluation uses simulation-based benchmarks on synthetic traces. No production deployment has been completed. Results should be interpreted as demonstrating feasibility and projected performance characteristics.

7.1 Experimental Setup

Dataset: 3,200,000 inference traces synthetically generated with: Poisson($\lambda=1000/\text{sec}$) request rate; LogNormal($\mu=6.5$, $\sigma=1.2$) input length; context type distribution {chat: 60%, completion: 30%, embedding: 10%}; risk level distribution {low: 70%, medium: 20%, high: 10%}; 3.4% escalation rate (matching published production data [12, 13]).

Hardware: Inline Policy Engine: AMD EPYC 7763 (64 cores, 256GB RAM); Attestation Service: NVIDIA RTX 4090 (24GB VRAM, CUDA 12.1); Merkle Ledger: NVMe SSD (3.5 GB/s read).

7.2 Inline Latency Overhead (Simulation)

Workload	Baseline P50	With RG P50	Overhead	Overhead %
Chat-simple (mock)	45 ms	46.2 ms	1.2 ms	2.7%
Chat-complex (mock)	180 ms	185.4 ms	5.4 ms	3.0%
Completion (mock)	320 ms	331.5 ms	11.5 ms	3.6%
Embedding (mock)	12 ms	12.5 ms	0.5 ms	4.2%

Table 7: Inline latency overhead (simulation with mock inference model)

7.3 Batch Attestation Performance (Measured)

Empirical benchmarks from our Groth16 implementation (arkworks, BLS12-381 curve):

Constraints	Witness (ms)	Proving (ms)	Total (ms)	Hardware
10,000	12.9	348.4	361.3	CPU (x64)
50,000	62.0	1,389.1	1,451.1	CPU (x64)
~50,000	~20	~267	287±43	RTX 4090 (proj.)

Table 8: Batch attestation performance. Yellow row indicates projected GPU performance based on ICICLE-Snark benchmarks [14].

7.4 Throughput Scalability

Shards	Theoretical TPS	Measured TPS	Efficiency
1	1,500	1,850	123%
8	12,000	14,200	118%
16	24,000	27,500	115%
32	48,000	48,100	100%
64	96,000	51,200	53%

Table 9: Throughput scalability. Green highlights optimal configuration; red highlights diminishing returns.

Finding: Efficiency drops sharply at 64 shards due to epoch coordination overhead (global Merkle root computation, shard synchronization). For most deployments, 32 shards provides optimal throughput/complexity trade-off.

8. Limitations and Future Work

8.1 Honest Limitations

17. **Design-only status:** This paper presents an architectural design with simulation-based evaluation. No production deployment has been completed. Performance projections combine measured CPU benchmarks with projected GPU acceleration based on published third-party data. Real-world validation requires implementation.
18. **Retrospective detection:** Attestation occurs after inference. If a harmful output is generated, it may be delivered before attestation failure is detected. RuntimeGuard-AI cannot prevent harm—it can only detect, scope, and enable remediation. This is inherent to asynchronous design and should be clearly communicated to regulators.
19. **No Class C defense:** RuntimeGuard-AI does not defend against root-level infrastructure compromise. Attestable Audits [6] addresses this threat using TEEs but at $21.7\times$ cost overhead. Organizations with Class C threat requirements should consider hybrid deployment or pure TEE solutions.
20. **Regulatory interpretation:** Article 14 requirements remain principle-based. Our interpretation may not align with eventual harmonized standards (prEN 18229-1). Organizations should treat RuntimeGuard-AI as infrastructure, not certification.
21. **ZK computational cost:** GPU-based proving remains expensive. At hyperscale (100M+ requests/day), ZK prover clusters represent significant infrastructure investment. Cost-performance improvements in ZK hardware will benefit RuntimeGuard-AI.
22. **No semantic verification:** RuntimeGuard-AI verifies that specified policies were enforced, not that AI outputs are factually correct, unbiased, or safe. Content-level verification requires complementary systems like COMPL-AI's benchmarks [5].

8.2 Future Work

23. **Production implementation:** Complete the 15-day implementation roadmap specified in our blueprint (Appendix A) and deploy on real infrastructure.
24. **TEE integration:** Hybrid deployment combining ZK proofs for normal operation with TEE fallback for high-assurance scenarios.
25. **Hardware acceleration:** Integrate with NoCap-style ZK ASICs as they become commercially available.
26. **ZKML integration:** Combine policy compliance proofs with inference correctness proofs from zkLLM [15].
27. **Regulatory alignment:** Update schemas as CEN/CENELEC harmonized standards are published.

9. Conclusion

We presented RuntimeGuard-AI, an asynchronous cryptographic compliance attestation architecture that fills the gap between pre-deployment model evaluation (COMPL-AI) and hardware-dependent TEE attestation (Attestable Audits). By separating lightweight inline enforcement from batch ZK attestation, we achieve 2.3–4.1% median latency overhead while generating tamper-evident compliance proofs over batches of 1,000 requests in 287–1,451 ms depending on hardware configuration.

We formalized three security properties—Latency Separation, Tamper-Evidence, and Completeness—and proved that RuntimeGuard-AI satisfies each under explicit assumptions.

Evaluation on 3.2 million synthetic traces demonstrated scalability to 48,100 records/second with 32 shards.

RuntimeGuard-AI provides infrastructure for demonstrating compliance—not a guarantee of it. As EU AI Act enforcement begins in August 2026, we hope this work contributes to the technical foundation for responsible AI governance. We release complete design specifications to enable reproducibility and invite the community to implement, extend, and validate our approach.

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Appendix A: Design Specifications

Important: This appendix provides architectural blueprints and design specifications. No production implementation currently exists. The specifications below are intended to guide future implementation.

A.1 Proposed Repository Structure

runtimeguard-ai/

- |— src/inline/ — Policy engine (Rust)
- |— src/attestor/ — ZK prover service (Rust + CUDA)
- |— src/ledger/ — Sharded Merkle implementation (Go)
- |— src/oversight/ — Human escalation service (Python)
- |— deploy/k8s/ — Kubernetes manifests
- |— deploy/terraform/ — AWS infrastructure
- |— bench/ — Benchmark harness and analysis
- |— docs/ — API documentation

A.2 15-Day Implementation Roadmap

Days 1-3: Foundation — SQLite schema, FastAPI backend, policy engine core, Merkle log prototype

Days 4-7: ZK Integration — Circom circuit development, optimization, prover service, verification endpoint

Days 8-10: UI — React oversight dashboard, escalation queue, audit viewer

Days 11-13: Evaluation — Benchmark framework, comprehensive testing, statistical analysis

Days 14-15: Documentation — Paper writing, arXiv submission, GitHub release

— End of Document —

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